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Stata Project Report

-Introduction to Capital Market Databases and Statistical Analysis Software

The Capital Asset Pricing Model (CAPM) is a model that explains the linear relationship between the expected excess return on an asset and its beta relative to the aggregate market portfolio. However, the traditional CAPM does not differentiate among the four covariation components, a combination of negative and positive market and asset returns (Bollerslev et al., 2022). The primary objective of this paper is to compute monthly realized semi-betas derived from daily stock returns. The following part will show the steps taken for data preprocessing, including collection and cleaning, summary statistics, regression results, and interpretations.

1 Extract the data from CRSP databse in WRDS

First, we extracted the daily data from the Center for Research in Securities Prices (CRSP) from January 1, 1963, to December 31, 2019. We included all stocks that have CRSP codes 10 and 11 by filtering the share code (shrcd). Share Class indicates the class of share, and we filtered it as "A" to extract each company's common stocks. We only selected stocks with active trading (trdstat) and regular way (secstat) status. Finally, as stated in the article, we removed all prices that were less than five dollars. We extracted the following variables: PERMNO, PERMCO, price, return, and vwretd. We used “vwretd” as the market return. The next figure illustrates each step taken in the CRSP database.

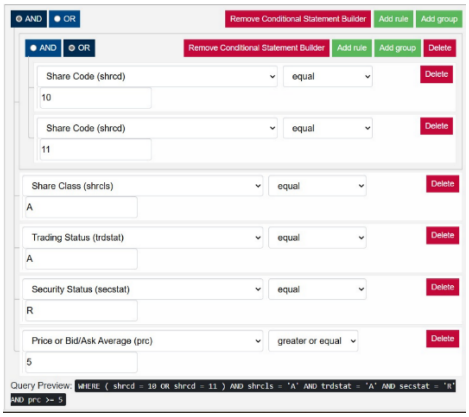


Figure 1. Filtering Process of the WRDS Database

2 Data Pre-Processing

- **Describing the Data**

After all filtration, CRSP provided us with a total of 2,046,068 daily level observations. At first glance, our RET mean value is .0008226 and it ranges from -.6799405 to 5.179978. Our value-weighted return (vwretd) has a mean value of .0004498 and ranges from -.1713457 to .1149183. Also, RET has a higher volatility with a standard deviation .0308692, compared to .0102866 (vwretd). Since we are working with panel data, we used the "xtset" command to organize our observations based on PERMCO and monthly date.

- **Checking for Duplicates**

```
. duplicates report

Duplicates in terms of all variables

--------------------------------------
   copies | observations       surplus
----------+---------------------------
        1 |      2046068             0
--------------------------------------
```

Our dataset contains no duplicate entries.

- **Validating the one-to-one mapping between PERMCO and PERMNO**

```
. tabulate unique_permno_count

unique_perm |
   no_count |      Freq.     Percent        Cum.
------------+-----------------------------------
          1 |  2,002,303       97.86       97.86
          2 |     26,631        1.30       99.16
          3 |      9,023        0.44       99.60
          6 |      8,111        0.40      100.00
------------+-----------------------------------
      Total |  2,046,068      100.00
```

We found that only 2,002,303 PERMCOs in the dataset are uniquely associated with one PERMNO. According to Zihan, this issue should not occur as we filtered by share code 10 or 11 and limited the share class to A. Despite multiple attempts, some PERMCOs still map to multiple PERMNOs (2, 3 & 6). Given the large size of the dataset, we decided to retain only the entries where PERMCO is uniquely associated with a single PERMNO.

- **Assessing the Normality of the Data Distribution**

In the histogram plot of RET (Figure 2), we observed more extreme values in the right tail, while the overall values are concentrated around zero. VWRETD values (Figure 3) are also mostly clustered around zero with a longer left tail.

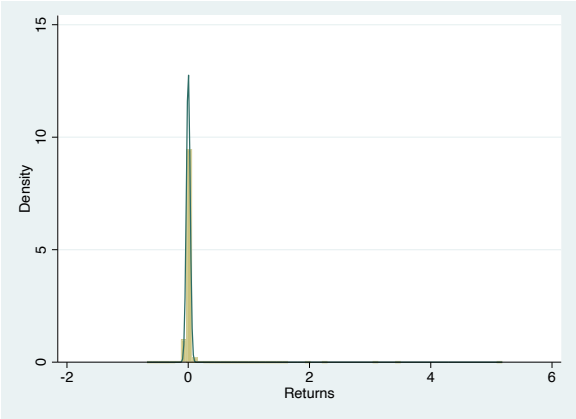


Figure 2. Return Distribution Graph winsorized) Code: hist RET, normal

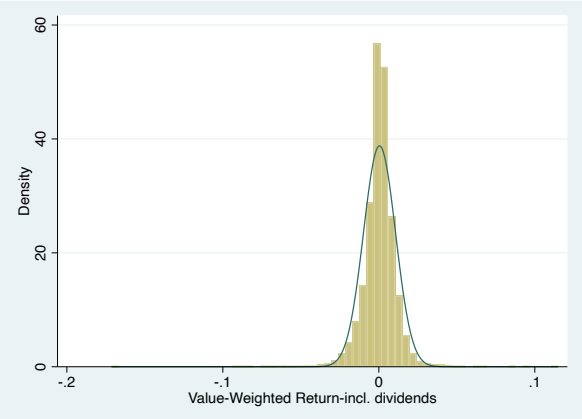


Figure 3. Market Return Distribution Graph (before (before winsorized) Code: hist vwretd normal

- **Dealing with Missing Values**

```
. misstable summarize

                                                     Obs<.
                                            +--------------------------------
             |                              | Unique
    Variable |     Obs=.     Obs>.     Obs<. | values        Min         Max
-------------+--------------------------------+-------------------------------
         RET |                786   2001517 |   >500  -.6799405    5.179978
-------------------------------------------------------------------------------
```

-Method: Cubic Spline Interpolation

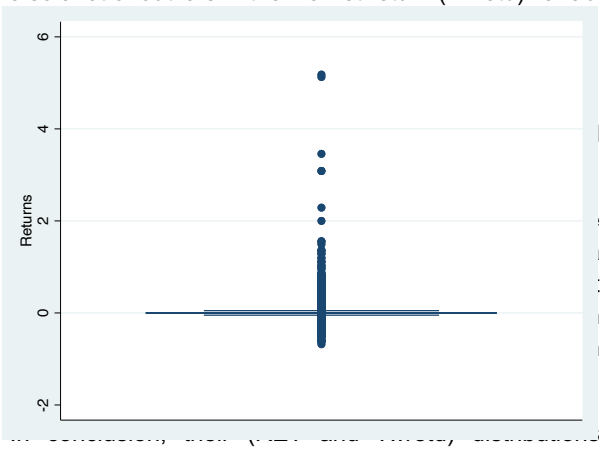
We observed that only the variable RET has missing values, with a total of 786 missing observations. We used cubic spline interpolation with the "csipolate" command to fill the missing RET values, as it provides smoother and

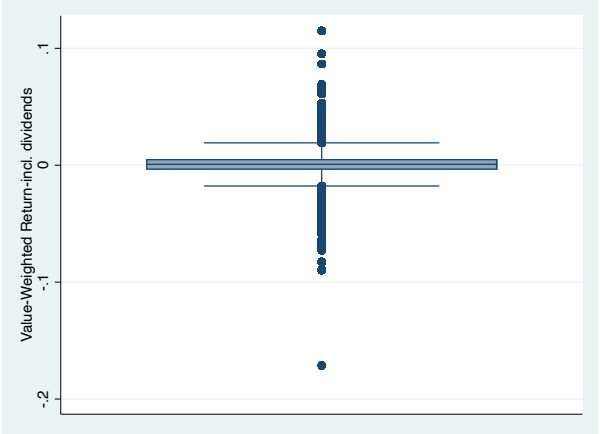
more accurate results compared to linear interpolation by fitting a cubic spline to the observed values. This method ensures continuity and captures the underlying trend in the data. And later we observed a slightly improved average adjusted R^2 comparing in the regression result without interpolation.

- **Dealing with Outlier**

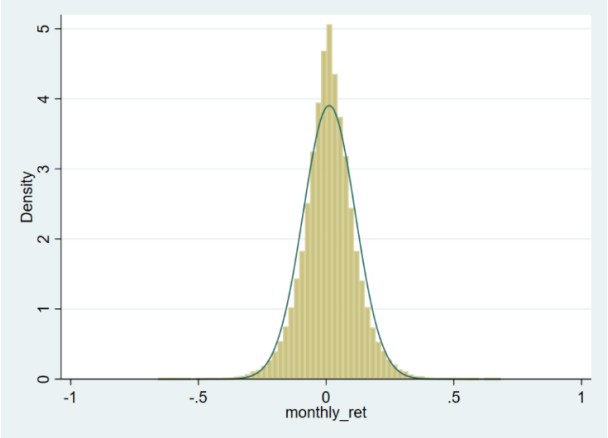
-Method: winsorization

In the graph box of RET (Figure 4), we observed numerous outliers, particularly on the positive side. There are also a lot of outliers in the market return (vwretd) variable, as can be seen in the graph box (Figure 5).





appear to be more normal (see Figure 6,7,8,9).



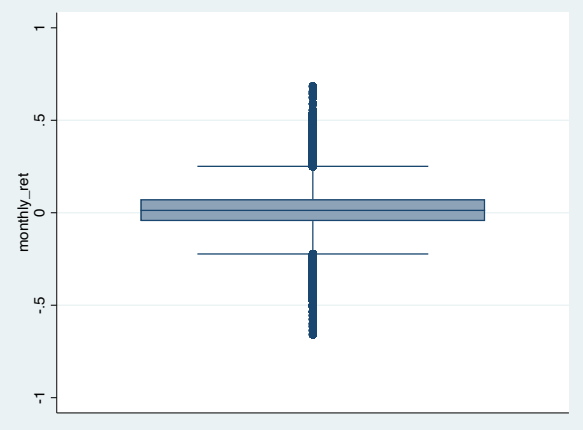
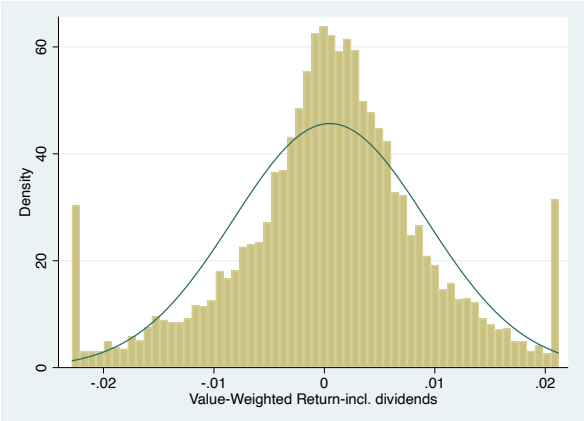


Figure 6. Histogram of MonthlyReturn

(aggregated montly after winsorization)

Figure 7. GraphBox of Monthly Return

(aggregated monthly winsorization)



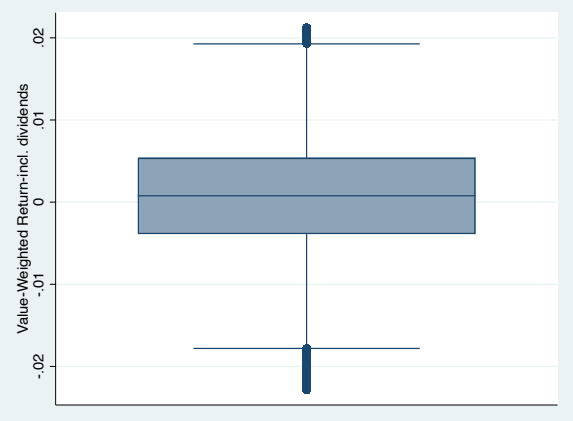


Figure 8. Histogram Graph of Market Return

Figure 9. Graph Box of Market Return

(after winsorization) (after winsorization)

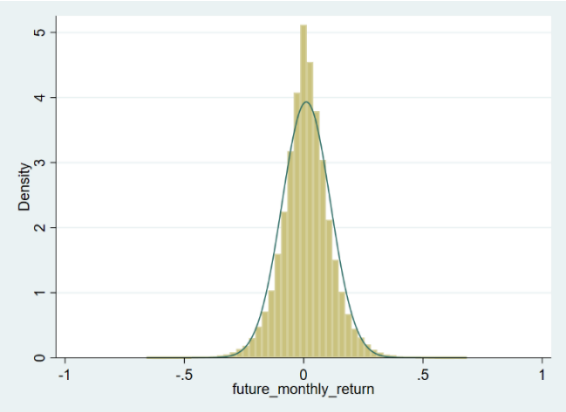


Figure 10. Future Monthly Return (after winsorization)

3 Calculate the Semi-betas

- **Performing Monthly Calculations for Semi-Beta Analysis**

Semi-betas were calculated by measuring the covariances between positive and negative stock returns and market returns, and then normalizing these covariances by the squared market returns for each stock and month. Based on the formula from the paper, we calculated the traditional beta (beta_trad).

- **Summary Statistics**

a) Pearson correlation check:

We first calculated the Pearson correlation coefficients between the variable beta_trad and its constituent variables (beta_n, beta_p, beta_mplus, and beta_mminus) and tested their statistical significance to examine the accuracy of the semi-betas' calculation.

b) Central tendency and variability check:

Second, we present the summary statistics of beta_trad and its components (beta_n, beta_p, beta_mplus, and beta_mminus), which reveal their central tendency (mean and median) and variability (standard deviation).

c) Result interpreting (see the figures below):

✧ **Version 1** (the original version that we analyse):

– **Version 1: Figure 1.a)** shows that beta_trad exhibited statistically significant linear correlations with some of its constituent variables ($p < 0.05$), while others were not significant. This suggests that our data may still contain inconsistencies or outliers, which could be affecting the accuracy of the semi-beta calculations.

– **Version 1: Figure 1.b)** shows large standard deviations (all standard deviations are larger then their means) further indicates the high variability in the data. These issues indicate that the analysis could be impacted by extreme values, and we need to address the outliers to maybe improve the reliability of the results.

⇨ **Conclusion**: we need to reexamine the caluculation of semi-betas!

To ensure the accuracy and reliability of our semi-betas calculations, we selected only months with at least 20 valid trading days. This threshold is based on the principle that a minimum of 20 days of data is needed to reduce estimation errors and capture the true relationship between assets and the market (Dimson, 1979). With fewer than 20 days of data, the sample size becomes too small, leading to potential inaccuracies in the covariance calculations and an increased risk of noise and outliers affecting the results. We specifically implemented this criterion by filtering out months with insufficient data, ensuring a robust and reliable analysis. We checked the results in Version 2 to confirm whether the results improved.

✧ **Version 2** (after filtering at least 20 valid trading days in one month):

- **Version 2: Figure 1.a)** shows that beta_trad exhibited statistically significant linear correlations with all constituent variables ($p < 0.05$), which is consistent with expectations.
- In **Version 2: Figure 1.b)**, we observe an improvement in the results. The summary statistics show a reduction in the mean and standard deviation across the variables compared to the previous version. For instance, beta_trad now has a mean of 0.9219817 and a standard deviation of 0.8720462, indicating less variability. Similarly, the other variables, such as beta_n, beta_p, beta_mplus, and beta_mminus, show more consistent results with lower standard deviations and means closer to their medians, suggesting that the data filtering has led to more reliable estimates.

```
. pwcorr beta_trad beta_n beta_p beta_mplus beta_mminus, sig star(0.05)

             | beta_t~d   beta_n   beta_p beta~lus beta~nus
-------------+---------------------------------------------
   beta_trad |   1.0000
             |
             |
      beta_n |   0.2036*  1.0000
             |   0.0000
             |
      beta_p |   0.7039* -0.0028   1.0000
             |   0.0000   0.3613
             |
  beta_mplus |  -0.2717*  0.0076*  0.0076*  1.0000
             |   0.0000   0.0147   0.0139
             |
 beta_mminus |  -0.6216* -0.0001   0.0000   0.0051   1.0000
             |   0.0000   0.9653   0.9909   0.1027
             |
```

Version 1: Figure 1.a)

```
. pwcorr beta_trad beta_n beta_p beta_mplus beta_mminus, sig star(0.05)

             | beta_t~d   beta_n   beta_p beta~lus beta~nus
-------------+---------------------------------------------
   beta_trad |   1.0000
             |
             |
      beta_n |   0.5528*  1.0000
             |   0.0000
             |
      beta_p |   0.6036*  0.1006*  1.0000
             |   0.0000   0.0000
             |
  beta_mplus |  -0.3780*  0.0348*  0.2434*  1.0000
             |   0.0000   0.0000   0.0000
             |
 beta_mminus |  -0.3758*  0.0517*  0.1277*  0.3687*  1.0000
             |   0.0000   0.0000   0.0000   0.0000
             |
```

Version 2: Figure 2.a)

```
. tabstat beta_trad beta_n beta_p beta_mplus beta_mminus, stats(mean median sd)

   stats |  beta_t~d    beta_n    beta_p  beta~lus  beta~nus
---------+--------------------------------------------------
    mean |   .8470989  .6438954  .9217951   .390027  .3285646
     p50 |   .8157173  .5214324  .6626909  .2003084  .1314794
      sd |   6.581502   1.36669  4.650522  1.813507  4.081605
-------------------------------------------------------------
```

Version 1: Figure 1.b)

```
. tabstat beta_trad beta_n beta_p beta_mplus beta_mminus, stats(mean median sd)

   stats |  beta_t~d    beta_n    beta_p  beta~lus  beta~nus
---------+--------------------------------------------------
    mean |   .9219817  .6263809  .8289513  .3129012  .2204494
     p50 |   .8905772  .5415946  .6864771  .1851982  .1244396
      sd |   .8720462   .448689  .6123754  .3888918  .2857596
-------------------------------------------------------------
```

Version 2: Figure 1.b)

In this latter version, we achieved very similar summary statistics to those in the article for our betas. For instance, our traditional beta has a mean of 0.92 (article: 0.99), a median of 0.89 (article: 0.92), and a standard deviation of 0.87 (article: 0.76). Also, our beta_n has a mean of 0.62 (article: 0.60), a median of 0.54 (article: 0.54), and a standard deviation of 0.44 (article: 0.36). Moreover, our correlation results follow the same pattern as the article results. For example, the correlation between beta_mminus and beta traditional is -0.37 (article: -0.29).

4 Fama-MacBeth Regression and Result Analysis

We have two versions of the regression: one that includes all stocks with return data, without any filtering, and another that filters the stocks, only including those with at least 20 valid trading days per month.

In the first version of our analysis, we performed two regressions. The first regression used future monthly returns as the dependent variable and four semi-betas as independent variables, with 101,550 observations, yielding an 8% R-squared but no significant results. The second regression used traditional beta with future returns, resulting in a lower R-squared of 2.5%, with no significant findings. Both regressions were conducted using the Fama-MacBeth method ("xtfmb").

Version 1

```
. xtfmb future_monthly_return beta_n beta_p beta_mplus beta_mminus

Fama-MacBeth (1973) Two-Step procedure      Number of obs     =    101550
                                            Num. time periods =       683
                                            F(  4,   682)     =      0.92
                                            Prob > F          =    0.4492
                                            avg. R-squared    =    0.0837
------------------------------------------------------------------------------
             |            Fama-MacBeth
future_mon~n |      Coef.   Std. Err.      t    P>|t|    [95% Conf. Interval]
-------------+----------------------------------------------------------------
      beta_n |   .0025971   .0017439     1.49   0.137    -.0008271    .0060212
      beta_p |    .000183    .001214     0.15   0.880    -.0022006    .0025666
 beta_mplus |  -.0024617   .0021421    -1.15   0.251    -.0066676    .0017442
beta_mminus |  -.0003392   .0023711    -0.14   0.886    -.0049948    .0043163
       _cons |   .0098379    .001751     5.62   0.000     .0063998    .0132759
------------------------------------------------------------------------------

. xtfmb future_monthly_return beta_trad

Fama-MacBeth (1973) Two-Step procedure      Number of obs     =    101550
                                            Num. time periods =       683
                                            F(  1,   682)     =      2.38
                                            Prob > F          =    0.1236
                                            avg. R-squared    =    0.0251
------------------------------------------------------------------------------
             |            Fama-MacBeth
future_mon~n |      Coef.   Std. Err.      t    P>|t|    [95% Conf. Interval]
-------------+----------------------------------------------------------------
   beta_trad |   .0009855   .0006392     1.54   0.124    -.0002696    .0022405
       _cons |   .0099378   .0015869     6.26   0.000      .006822    .0130536
------------------------------------------------------------------------------
```

Version 2

```
. xtfmb future_monthly_return beta_n beta_p beta_mplus beta_mminus

Fama-MacBeth (1973) Two-Step procedure      Number of obs     =     59843
                                            Num. time periods =       521
                                            F(  4,   520)     =      2.23
                                            Prob > F          =    0.0643
                                            avg. R-squared    =    0.1473
------------------------------------------------------------------------------
             |            Fama-MacBeth
future_mon~n |      Coef.   Std. Err.      t    P>|t|    [95% Conf. Interval]
-------------+----------------------------------------------------------------
      beta_n |  -.0153146   .0169213    -0.91   0.366    -.0485572    .0179279
      beta_p |   .0047338   .0046433    -1.02   0.308    -.0138557    .0043881
 beta_mplus |   .0163875   .0222944     0.74   0.463    -.0274106    .0601856
beta_mminus |  -.0138944   .0060906    -2.28   0.023    -.0258595   -.0019293
       _cons |   .0227105   .0098975     2.29   0.022     .0032665    .0421544
------------------------------------------------------------------------------

. xtfmb future_monthly_return beta_trad

Fama-MacBeth (1973) Two-Step procedure      Number of obs     =     59843
                                            Num. time periods =       521
                                            F(  1,   520)     =      0.82
                                            Prob > F          =    0.3645
                                            avg. R-squared    =    0.0449
------------------------------------------------------------------------------
             |            Fama-MacBeth
future_mon~n |      Coef.   Std. Err.      t    P>|t|    [95% Conf. Interval]
-------------+----------------------------------------------------------------
   beta_trad |    .003359   .0037006     0.91   0.364     -.003911    .0106289
       _cons |   .0119138   .0029408     4.05   0.000     .0061365    .0176911
------------------------------------------------------------------------------
```

In the second version of our analysis, we performed two regressions using a filtered sample of 59,843 observations, based on the requirement of at least 20 valid monthly trading days. The first regression, with future monthly returns as the dependent variable and four semi-betas as independent variables, resulted in a 14% R-squared. We found one significant result for beta_mminus, which had a negative coefficient of -1.3% and a p-value of 0.023. This suggests that a 1 standard deviation increase in beta_mminus is associated with a 1.3% decrease in future returns. The second regression, using traditional beta, had a lower R-squared of 4.4% and no significant findings. Both regressions were conducted using the Fama-MacBeth method ("xtfmb").

To conclude, the article finds that semi-betas resulting from the covariation of negative market and negative asset returns predict significantly higher future returns, while semi-betas derived from the covariation of negative market and positive asset returns predict significantly lower future returns (Bollerslev et al., 2022). Our semi-beta beta_mminus, which represents the covariation of negative market and positive asset returns, predicts significantly (p: 0.023, < 0.05) lower future returns (-1.3%), in line with the article's results. Moreover, there are no statistically significant results for "beta_p" and "beta_mplus", both in the article and in our regression results. The only distinction is that the article reported significant results concerning the covariation between negative market returns and negative asset returns (beta_n) semi-beta. This could be due to the fact that we are unsure how the data was processed by the researchers, and it is possible that the data has been updated since June 2021.

- **Annualized Risk Premium**

In the first version of our analysis, we used the Newey-West robust standard errors (with 10 lags) to calculate the annualized risk premiums for the four semi-betas. The results are as follows:

β_n	β_p	β_{m+}	β_{m-}
.03116484	.00219591	-.02954023	-.00407078

β_n	β_p	β_{m+}	β_{m-}
-.18377548	-.05680544	.19664986	-.16673294

The results showed no significant coefficients for most semi-betas. However, was statistically significant in the second version with a negative coefficient of -16.67%, indicating that a one standard deviation increase in is associated with a 16.67% decrease in the annualized risk premium. This suggests that market downturns negatively affect future returns, highlighting the impact of market volatility.

-References-

Bollerslev, T., Patton, A. J., & Quaedvlieg, R. (2022). Realized semibetas: Disentangling “good” and “bad” downside risks. Journal of Financial Economics, 144(1), 227-246.

Dimson, E. (1979). "Risk Measurement When Shares are Subject to Infrequent Trading", Journal of Financial Economics, 7(2): 197-226.